TP4 : Bases d'Ansible (inventaire, playbooks, clés SSH, rôles)

Partie _0 : Préparation de l'inventaire : [Sur la VM Contrôle - 192.168.159.20]

1. installe sshpass :

```
arij@arij:~$ sudo apt install -y ansible sshpass
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
ansible est déjà la version la plus récente (9.2.0+dfsg-0ubuntu5).
Les NOUVEAUX paquets suivants seront installés :
  sshpass
0 mis à jour, 1 nouvellement installés, 0 à enlever et 76 non mis à jour.
Il est nécessaire de prendre 11,7 ko dans les archives.
Après cette opération, 35,8 ko d'espace disque supplémentaires seront utilisés.
Réception de :1 http://tn.archive.ubuntu.com/ubuntu noble/universe amd64 sshpass amd64 1.09-1 [11,7 kB]
11,7 ko réceptionnés en 0s (30,0 ko/s)
Sélection du paquet sshpass précédemment désélectionné.
(Lecture de la base de données... 147224 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../sshpass_1.09-1_amd64.deb ...
Dépaquetage de sshpass (1.09-1) ...
Paramétrage de sshpass (1.09-1) ...
Traitement des actions différées (« triggers ») pour man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

2.vérifier

```
arij@arij:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/arij/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/arij/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Aug 31 2026, 10:18:26) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
```

3.Créer le fichier inventory :

```
  GNU nano 7.2                                          inventory.txt *
[webserver]
192.168.159.22

[ftpserver]
192.168.159.23

[nodes:children]
webserver
ftpserver

[nodes:vars]
ansible_user=arij
```

4. Vérifie l'inventaire

```
arij@arij:~/ansible/demo1$ ansible-inventory -i inventory.txt --graph
@all:
  |--@ungrouped:
  |--@nodes:
  |  |--@webserver:
  |  |  |--192.168.159.22
  |  |--@ftpserver:
  |  |  |--192.168.159.23
```

5. Ajouter les empreintes des machines

```
arij@arij:~/ansible/demo1$ ssh-keyscan -H 192.168.159.22 >> ~/.ssh/known_hosts
# 192.168.159.22:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.20
# 192.168.159.22:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.20
# 192.168.159.22:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.20
# 192.168.159.22:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.20
# 192.168.159.22:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.20
arij@arij:~/ansible/demo1$ ssh-keyscan -H 192.168.159.23 >> ~/.ssh/known_hosts
# 192.168.159.23:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.20
# 192.168.159.23:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.20
# 192.168.159.23:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.20
# 192.168.159.23:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.20
# 192.168.159.23:22 SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.20
```

5. lancer le Ping Ansible

```
arij@arij:~/ansible/demo1$ ansible all -i inventory.txt -m ping -k
SSH password:
192.168.159.22 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
192.168.159.23 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Partie_1 : Ajouter un fichier ansible.cfg

1. Créer le fichier ansible.cfg

```
 GNU nano 7.2                                                ansible.cfg *
[defaults]
inventory = ./inventory.txt
remote_user = arij
host_key_checking = False
deprecation_warnings = False
```

2. Tester :

```
arij@arij:~/ansible/demo1$ ansible all -m ping -k
SSH password:
192.168.159.22 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
192.168.159.23 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Partie_2 : Créer un premier playbook :

1.Créer fichier index.html

```
 GNU nano 7.2                                                index.html *
<!doctype html>
<html lang="fr">
<head>
  <meta charset="utf-8">
  <title>Démo Ansible</title>
</head>
<body>
  <h1>Bonjour Master DevOps & Cloud</h1>
</body>
</html>
```

2. Créer le playbook webserver_install.yml

```
 GNU nano 7.2                                          webserver_install.yml *
---
- name: Installer et configurer le serveur web
  hosts: webserver          # groupe de l'inventaire ("all" pour cibler tous les hôtes)
  become: true              # exécuter les tâches avec sudo
  gather_facts: true

  tasks:
    - name: Créer le dossier de test
      ansible.builtin.file:
        path: "/home/{{ ansible_user }}/test"
        state: directory
        owner: "{{ ansible_user }}"
        mode: '0755'

    - name: Installer Apache
      ansible.builtin.apt:
        name: apache2
        state: present
        update_cache: true

    - name: Copier le site web sur les nœuds
      ansible.builtin.copy:
        src: index.html
        dest: /var/www/html/index.html
        owner: www-data
        group: www-data
        mode: '0644'

    - name: Vérifier qu'Apache est démarré et activé au boot
```

3.Vérifier la syntaxe (Syntax-check) :

```
arij@arij:~/ansible/demo1$ ansible-playbook webserver_install.yml --syntax-check

playbook: webserver_install.yml
```

4. Faire un essai à blanc (Dry-run) :

```
arij@arij:~/ansible/demo1$ ansible-playbook webserver_install.yml -k -K --check --diff
SSH password:
BECOME password[defaults to SSH password]:

PLAY [Installer et configurer le serveur web] **********************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [192.168.159.22]

TASK [Créer le dossier de test] ************************************************************************************************
ok: [192.168.159.22]

TASK [Installer Apache] ********************************************************************************************************
ok: [192.168.159.22]

TASK [Copier le site web sur les nœuds] ****************************************************************************************
ok: [192.168.159.22]

TASK [Vérifier qu'Apache est démarré et activé au boot] ************************************************************************
ok: [192.168.159.22]

PLAY RECAP *********************************************************************************************************************
192.168.159.22             : ok=5    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

5. Exécution réelle :

```
arij@arij:~/ansible/demo1$ ansible-playbook webserver_install.yml -k -K
SSH password:
BECOME password[defaults to SSH password]:

PLAY [Installer et configurer le serveur web] **********************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [192.168.159.22]

TASK [Créer le dossier de test] ************************************************************************************************
ok: [192.168.159.22]

TASK [Installer Apache] ********************************************************************************************************
ok: [192.168.159.22]

TASK [Copier le site web sur les nœuds] ****************************************************************************************
ok: [192.168.159.22]

TASK [Vérifier qu'Apache est démarré et activé au boot] ************************************************************************
ok: [192.168.159.22]

PLAY RECAP *********************************************************************************************************************
192.168.159.22             : ok=5    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

6. Vérifie le site web

```
arij@arij:~/ansible/demo1$ curl http://192.168.159.22
<!doctype html>
<html lang="fr">
<head>
  <meta charset="utf-8">
  <title>Démo Ansible</title>
</head>
<body>
  <h1>Bonjour Master DevOps & Cloud</h1>
</body>
</html>
```

Partie_3 :Utiliser une paire de clés SSH privée/publique :

1. Générer une paire de clés SSH

```
arij@arij:~/ansible/demo1$ ssh-keygen -t ed25519 -C "ansible-control"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/arij/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/arij/.ssh/id_ed25519
Your public key has been saved in /home/arij/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:lmjq1Y6CGjUFled827SPFCFPFLg+igAB6F1u5GWKGaE ansible-control
The key's randomart image is:
+--[ED25519 256]--+
|+ .oo.   oo.     |
|o .o.o.oo o      |
|.E. X++  = .     |
|.. = =o.o.+      |
| .o . ooS+ o     |
| ... o o+ +      |
|.  o.....o o     |
| ...o..o  . .    |
|..  ...  .       |
+----[SHA256]-----+
```

2.Copier la clé publique sur les 2 nœuds gérés :

```
arij@arij:~/ansible/demo1$ ssh-copy-id arij@192.168.159.22
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/arij/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
arij@192.168.159.22's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'arij@192.168.159.22'"
and check to make sure that only the key(s) you wanted were added.

arij@arij:~/ansible/demo1$ ssh-copy-id arij@192.168.159.23
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/arij/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
arij@192.168.159.23's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'arij@192.168.159.23'"
and check to make sure that only the key(s) you wanted were added.
```

3. Tester le playbook SANS le -k :

```
arij@arij:~/ansible/demo1$ ansible-playbook webserver_install.yml -K
BECOME password:

PLAY [Installer et configurer le serveur web] ***************************************************************************

TASK [Gathering Facts] ***************************************************************************************************
ok: [192.168.159.22]

TASK [Créer le dossier de test] ******************************************************************************************
ok: [192.168.159.22]

TASK [Installer Apache] **************************************************************************************************
ok: [192.168.159.22]

TASK [Copier le site web sur les nœuds] **********************************************************************************
ok: [192.168.159.22]

TASK [Vérifier qu'Apache est démarré et activé au boot] ******************************************************************
ok: [192.168.159.22]

PLAY RECAP ***************************************************************************************************************
192.168.159.22             : ok=5    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Partie_4 : Playbook d'installation de conteneurs Docker :

1. Installer la collection Docker pour Ansible

```
arij@arij:~/ansible/demo1$ ansible-galaxy collection install community.docker
Starting galaxy collection install process
Nothing to do. All requested collections are already installed. If you want to reinstall them, consider using `--force`.
arij@arij:~/ansible/demo1$
```

2.Créer le Dockerfile :

```
 GNU nano 7.2                                    Dockerfile *
FROM nginx:alpine
COPY index.html /usr/share/nginx/html/index.html
```

3. Créer le docker-compose.yml

```
 GNU nano 7.2                                  docker-compose.yml *
services:
  web:
    build: .
    ports:
      - "8082:80"
    restart: always
```

3. Créer le playbook docker_install.yml

```
 GNU nano 7.2                                        docker_install.yml *
---
- name: Installer Docker, Jenkins, Portainer et le conteneur web
  hosts: nodes
  become: true
  tasks:
    - name: Installer les prérequis et Docker
      ansible.builtin.apt:
        name:
          - docker.io
          - docker-compose-v2
          - curl
        state: present
        update_cache: true

    - name: Démarrer Docker
      ansible.builtin.service:
        name: docker
        state: started
        enabled: true

    - name: Ajouter l'utilisateur au groupe docker
      ansible.builtin.user:
        name: "{{ ansible_user }}"
        groups: docker
        append: true

    - name: Lancer Portainer
      community.docker.docker_container:
        name: portainer
```

4. Exécuter le playbook Docker

```
arij@arij:~/ansible/demo1$ ansible-playbook docker_install.yml -K
BECOME password:

PLAY [Installer Docker, Jenkins, Portainer et le conteneur web] ****************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [192.168.159.22]
ok: [192.168.159.23]

TASK [Installer les prérequis et Docker] ***************************************************************************************
ok: [192.168.159.23]
ok: [192.168.159.22]

TASK [Démarrer Docker] *********************************************************************************************************
ok: [192.168.159.22]
ok: [192.168.159.23]

TASK [Ajouter l'utilisateur au groupe docker] **********************************************************************************
ok: [192.168.159.22]
ok: [192.168.159.23]

TASK [Lancer Portainer] ********************************************************************************************************
ok: [192.168.159.22]
ok: [192.168.159.23]

TASK [Lancer Jenkins (en conteneur Docker)] ************************************************************************************
ok: [192.168.159.22]
ok: [192.168.159.23]

TASK [Créer le dossier de l'application] ***************************************************************************************
ok: [192.168.159.23]
ok: [192.168.159.22]
```

```
TASK [Copier le Dockerfile, l'index.html et le compose] ************************************************************************
ok: [192.168.159.22] => (item=Dockerfile)
ok: [192.168.159.23] => (item=Dockerfile)
ok: [192.168.159.22] => (item=index.html)
ok: [192.168.159.23] => (item=index.html)
changed: [192.168.159.22] => (item=docker-compose.yml)
changed: [192.168.159.23] => (item=docker-compose.yml)

TASK [Lancer le conteneur web (build et up)] ***********************************************************************************
changed: [192.168.159.23]
changed: [192.168.159.22]

PLAY RECAP *********************************************************************************************************************
192.168.159.22             : ok=9    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
192.168.159.23             : ok=9    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

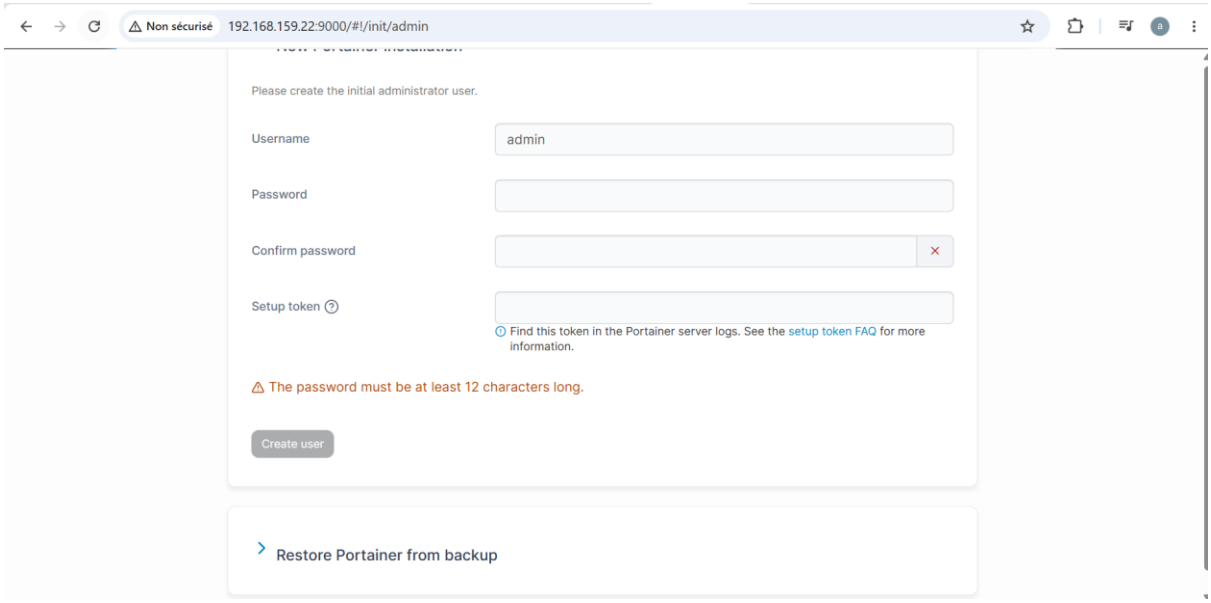
5. Vérifier sur ton navigateur Windows

Le Site web



Bonjour Master DevOps & Cloud

Portainer :



Jenkins



Partie_5 : Les Rôles :

1. Créer l'arborescence des rôles

```
arij@arij:~/ansible/demo1$ mkdir -p roles
ansible-galaxy role init roles/docker_portainer_install
ansible-galaxy role init roles/jenkins_install
ansible-galaxy role init roles/container_deployment
- Role roles/docker_portainer_install was created successfully
- Role roles/jenkins_install was created successfully
- Role roles/container_deployment was created successfully
```

2. Remplir le rôle docker_portainer_install

```
 GNU nano 7.2                                roles/docker_portainer_install/tasks/main.yml *
---
- name: Installer les prérequis et Docker
  ansible.builtin.apt:
    name:
      - docker.io
      - docker-compose-v2
      - curl
    state: present
    update_cache: true

- name: Démarrer Docker
  ansible.builtin.service:
    name: docker
    state: started
    enabled: true

- name: Ajouter l'utilisateur au groupe docker
  ansible.builtin.user:
    name: "{{ ansible_user }}"
    groups: docker
    append: true

- name: Lancer Portainer
  community.docker.docker_container:
    name: portainer
    image: portainer/portainer-ce:latest
    ports:
      - "9000:9000"
    volumes:
```

2. Remplir le rôle jenkins_install

```
 GNU nano 7.2                                roles/jenkins_install/tasks/main.yml *
---
- name: Lancer Jenkins (en conteneur Docker)
  community.docker.docker_container:
    name: jenkins
    image: jenkins/jenkins:lts
    ports:
      - "8081:8080"
    volumes:
      - jenkins_home:/var/jenkins_home
    restart_policy: always
```

3.Remplir le rôle container_deployment

```
 GNU nano 7.2                                roles/container_deployment/tasks/main.yml *
---
- name: Créer le dossier de l'application
  ansible.builtin.file:
    path: "/home/{{ ansible_user }}/webapp"
    state: directory
    owner: "{{ ansible_user }}"
    mode: '0755'

- name: Copier le Dockerfile, l'index.html et le compose
  ansible.builtin.copy:
    src: "{{ item }}"
    dest: "/home/{{ ansible_user }}/webapp/{{ item }}"
    owner: "{{ ansible_user }}"
  loop:
    - Dockerfile
    - index.html
    - docker-compose.yml

- name: Lancer le conteneur web (build et up)
  ansible.builtin.command:
    cmd: docker compose up -d --build
    chdir: "/home/{{ ansible_user }}/webapp"
  changed_when: false   # <--- AJOUTE CETTE LIGNE
```

5. Créer le playbook principal site.yml

```
 GNU nano 7.2                                          site.yml *
---
- name: Déploiement complet avec rôles
  hosts: nodes
  become: true
  roles:
    - docker_portainer_install
    - jenkins_install
    - container_deployment
```

6.Tester le playbook à base de rôles

```
arij@arij:~/ansible/demo1$ ansible-playbook site.yml -K
BECOME password:

PLAY [Déploiement complet avec rôles] ******************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.159.22]
ok: [192.168.159.23]

TASK [docker_portainer_install : Installer les prérequis et Docker] ************************************************
ok: [192.168.159.22]
ok: [192.168.159.23]

TASK [docker_portainer_install : Démarrer Docker] ******************************************************************
ok: [192.168.159.23]
ok: [192.168.159.22]

TASK [docker_portainer_install : Ajouter l'utilisateur au groupe docker] *******************************************
ok: [192.168.159.23]
ok: [192.168.159.22]

TASK [docker_portainer_install : Lancer Portainer] *****************************************************************
ok: [192.168.159.22]
ok: [192.168.159.23]

TASK [jenkins_install : Lancer Jenkins (en conteneur Docker)] ******************************************************
ok: [192.168.159.22]
ok: [192.168.159.23]

TASK [container_deployment : Créer le dossier de l'application] ****************************************************
ok: [192.168.159.23]
ok: [192.168.159.22]
```

```
TASK [container_deployment : Créer le dossier de l'application] ****************************************************
ok: [192.168.159.23]
ok: [192.168.159.22]

TASK [container_deployment : Copier le Dockerfile, l'index.html et le compose] *************************************
ok: [192.168.159.23] => (item=Dockerfile)
ok: [192.168.159.22] => (item=Dockerfile)
ok: [192.168.159.23] => (item=index.html)
ok: [192.168.159.22] => (item=index.html)
ok: [192.168.159.22] => (item=docker-compose.yml)
ok: [192.168.159.23] => (item=docker-compose.yml)

TASK [container_deployment : Lancer le conteneur web (build et up)] ************************************************
ok: [192.168.159.23]
ok: [192.168.159.22]

PLAY RECAP *********************************************************************************************************
192.168.159.22             : ok=9    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
192.168.159.23             : ok=9    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```